

Incremental Consistent k -Center Clustering

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k -Center Clustering

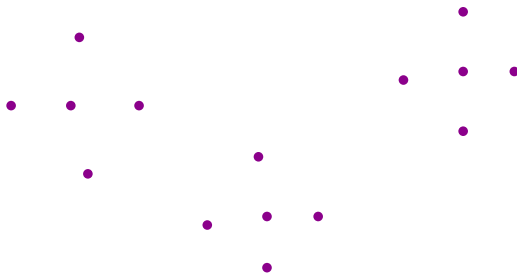
k -Center

Given: Point Set P in a metric space, integer $k \geq 1$

Goal: Find set $S \subseteq P$ s.t.

$$|S| \leq k$$

$\max_{p \in P} \text{dist}(p, S)$ is minimized.



$k = 3$

k -Center Clustering

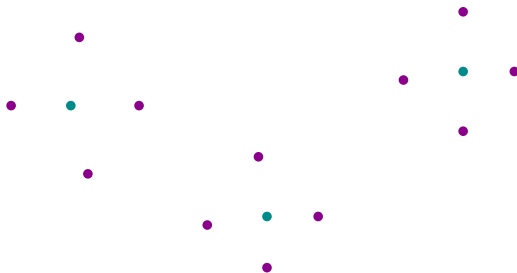
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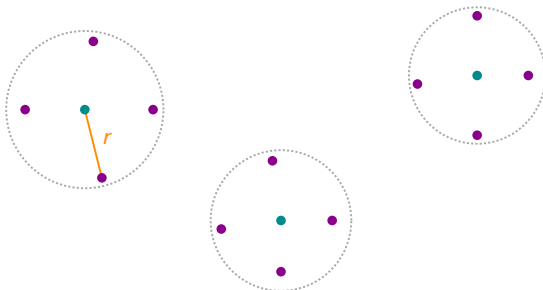
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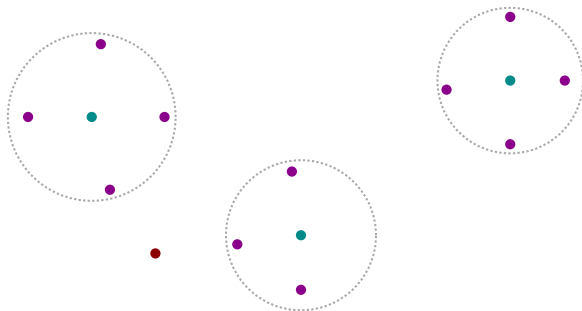
$\max_{p \in P} \text{dist}(p, S)$ is minimized.

NP-Hard to approximate for a factor of $2 - \varepsilon$

Incremental Consistent k -Center

→ Allow insertions to P

New Goal: maintain S under these updates

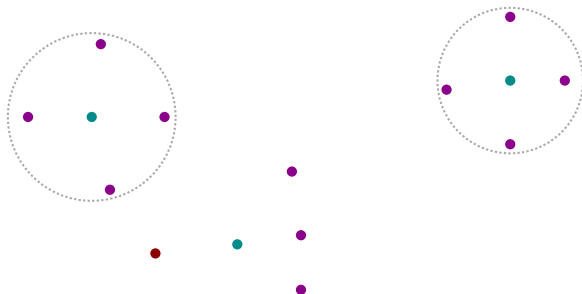


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Incremental Consistent k -Center

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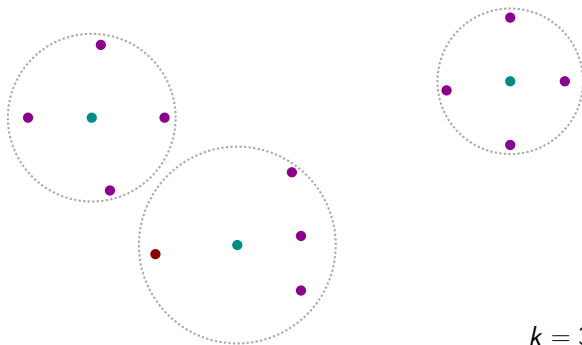


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Incremental Consistent k -Center

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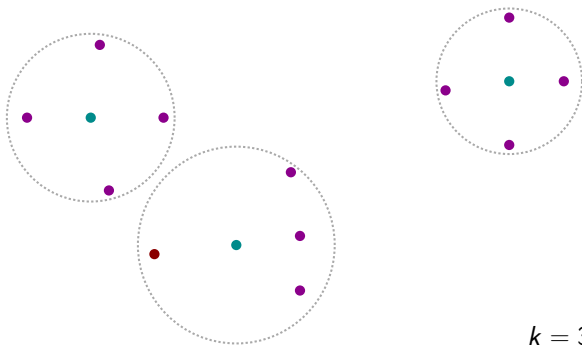


Recurse: Number of changes to S

Incremental Consistent k -Center

→ Allow insertions to P

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Optimize: $O(1)$ approximation, $O(1)$ recourse

Related Work

- Consistent clustering introduced [Lattanzi, Vassilvitskii 2017]
 - Upper bound $O(1)$ -approximation: $O(k^2 \log n)$ total recourse
 - Lower bound: $\Omega(k \log n)$ total recourse
- Recent results for k -median focusing on optimal recourse:
 - [Guo, Kulkarni, Li and Xian 2021]
 - [Bhattacharya, Costa and Farokhnejad 2024]
 - [Bhattacharya, Costa, Garg, Lattanzi and Parotsidis 2024]
- Consistent k -center:
 - [Charikar, Chekuri, Feder and Motwani 1997]
 - 8-approximation
 - [Forster and Skarlatos 2025]
 - Fully dynamic k -center with recourse 1: 50-approximation
 - Incremental/Decremental k -center with recourse 1: 6-approximation

Our Results

Theorem

There is a deterministic incremental algorithm for the k -center clustering problem that maintains a set of centers s.t.:

- Solution is a $(3 + \sqrt{5})$ -approximate,
- Worst-case recourse is at most 1,
- Worst-case update time is $O(k^2)$.

Worst-case lower bound:

Theorem

Consider a point set P in a metric space subject to point insertions, integer $k \geq 1$, and approximation factor $\alpha \in [1, \sqrt{2})$. Any partially dynamic α -approximate k -center algorithm has a worst-case recourse of k .

Estimating the Radius

Similar to [Forster and Skarlatos 2025]

Hochbaum and Schmoys 1986

If there are $k + 1$ points in P with pairwise distances $\geq r$ then $r \leq 2 \cdot R^*$.

Idea: k points from S + one point from $P \setminus S$ as certificate

Once pairwise distances increase by factor $\alpha > 1$

→ Increase estimate of radius r'

→ The time that r' remains unchanged is a **phase**

Primary Objective: Alter S such that **distances** between points in S are increased

Types of Centers

Leader centers S_L

Center c in S is a **leader center** iff there is no other center $c' \in S \setminus \{c\}$ s.t. $\text{dist}(c, c') < \alpha r'$.

Other centers form the **regular** centers S_R

Idea: Only move regular centers to get closer to primary objective

Initially: Leader centers $\rightarrow$ responsible for points up to distance $\alpha r'$

Regular centers $\rightarrow$ responsible up to a distance $\beta r'$ for suitable $\alpha > \beta \geq 1$

During a Phase - Easy Update

Responsibilities do not change:

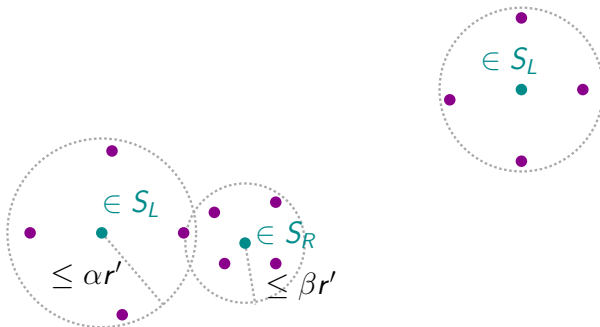
- $\text{dist}(p^+, S_L) \leq \alpha r'$ or
- $\text{dist}(p^+, S_R) \leq \beta r'$



→ Add p^+ to cluster

During a Phase - Example Update

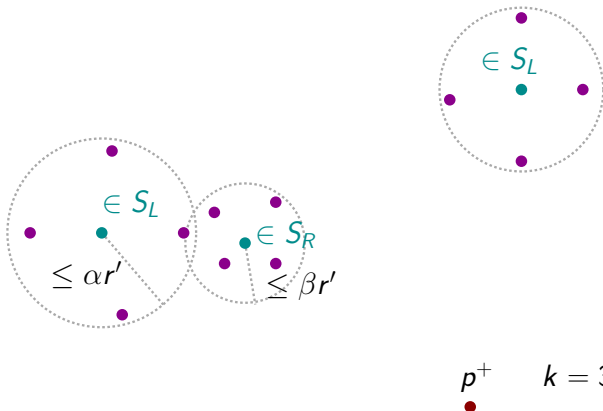
r' ... current estimate for radius



$$k = 3$$

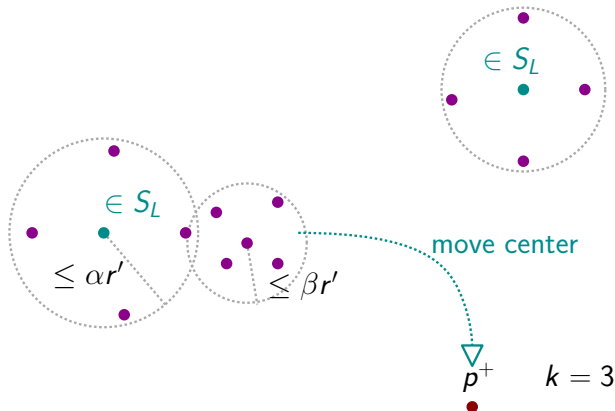
During a Phase - Example Update

r' ... current estimate for radius



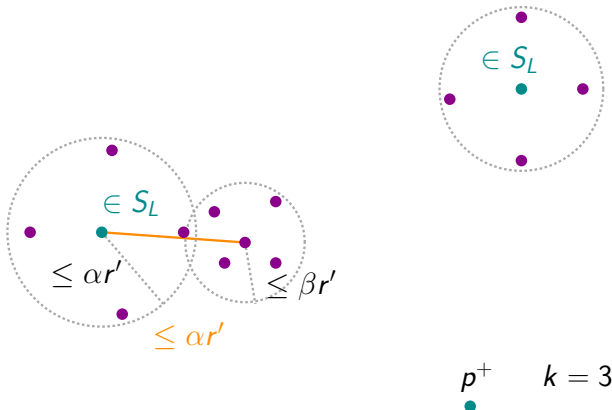
During a Phase - Example Update

r' ... current estimate for radius



During a Phase - Example Update

r' ... current estimate for radius



→ Leader center responsible for points at distances up to $\alpha r' + \beta r'$

Invariants

Invariant 1: $|S| = k$

Invariant 2: There is $p_S \in P \setminus S$ s.t. pairwise distances of points in $S \cup \{p_S\}$ are $\geq r'$.

Invariant 3: For every center $c_1 \in S_R$, there is center $c_2 \in S_L$ s.t. $\text{dist}(c_1, c_2) \leq \alpha r'$.

Invariant 4: Every point $p \in P$ is within distance $\alpha r' + \beta r'$ from its closest center in S .

Parameters: $\alpha = \frac{3+\sqrt{5}}{2}$ and $\beta = \frac{1+\sqrt{5}}{2}$

→ Directly would give $(4 + 2\sqrt{5})$ -approximation

Algorithm

Gonzalez Operation

At the beginning of phase order center points:

- Choose arbitrary center as c_1
- c_i maximizes $\text{dist}(c_i, \{c_1, c_2, \dots, c_{i-1}\})$

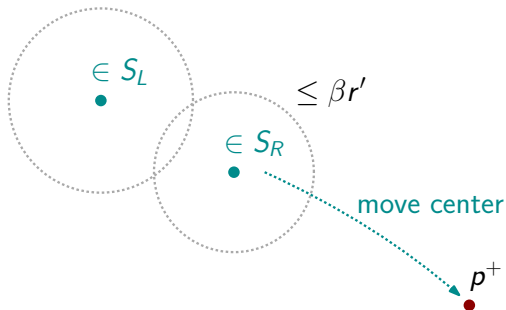
If p^+ is inserted, s.t. $\text{dist}(p^+, S_L) > \alpha r'$ and $\text{dist}(p^+, S_R) > \alpha r'$:

Search for the regular center $c_i \in S_R$ (with highest index)
minimizing $(c_i, S \setminus \{c_i\})$

→ move center from that point

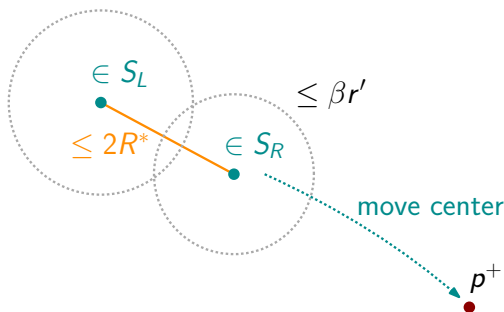
More Detailed Analysis - Leader Case

Assume: $\text{dist}(p^+, S) \geq \text{smallest pairwise distance in } S$



More Detailed Analysis - Leader Case

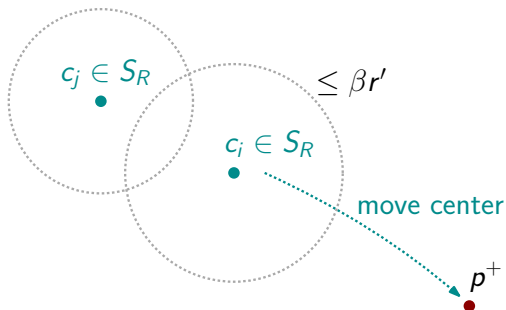
Assume: $\text{dist}(p^+, S) \geq$ smallest pairwise distance in S



$$\rightarrow \text{dist}(p, S) \leq 2R^* + \beta r' \leq 2R^* + 2\beta R^* = (3 + \sqrt{5})R^*$$

More Detailed Analysis - Regular Case

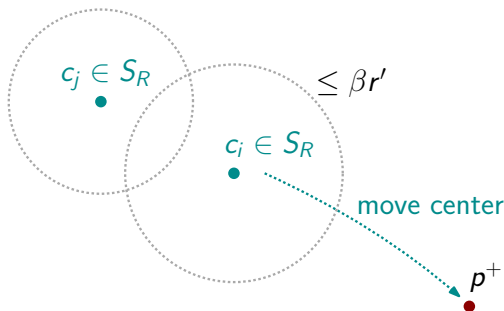
Smallest pairwise distance in S between two regular centers



What happens when c_j also moves?

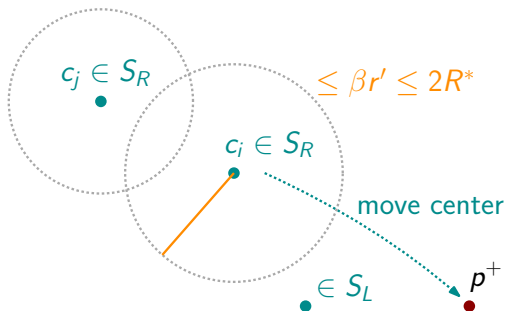
More Detailed Analysis - Regular Case

Case 1: Smallest pairwise distance in S is at least $\beta r' \rightarrow k + 1$ point certificate



More Detailed Analysis - Regular Case

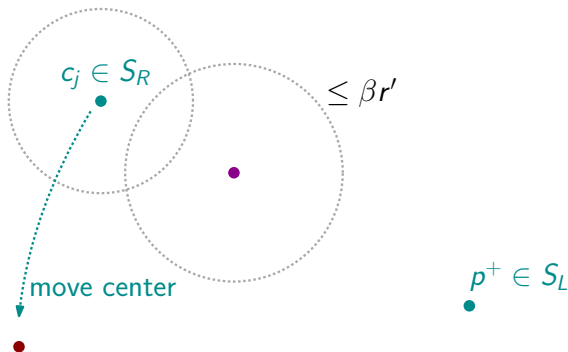
Case 1: Smallest pairwise distance in S is at least $\beta r' \rightarrow k + 1$ point certificate



$$\rightarrow \text{dist}(p, S) \leq 2R^* + \alpha r' \leq 2R^* + \frac{2\alpha}{\beta} R^* = (3 + \sqrt{5})R^*$$

More Detailed Analysis - Regular Case

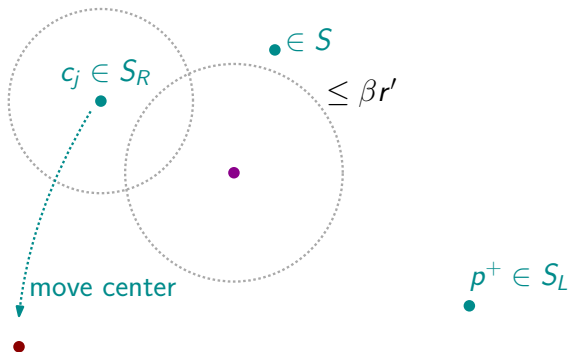
Case 2: Smallest pairwise distance in S is at most $\beta r'$
Consider the update, when center at c_j moves:



From Gonzalez: $\text{dist}(c_i, S) \leq \text{dist}(c_j, S \setminus \{c_j\}) \leq 2R^*$

More Detailed Analysis - Regular Case

Case 2: Smallest pairwise distance in S is at most $\beta r'$
Consider the update, when center at c_j moves:



From Gonzalez: $\text{dist}(c_i, S) \leq \text{dist}(c_j, S \setminus \{c_j\}) \leq 2R^*$
→ Triangle inequality: $(3 + \sqrt{5})R^*$

Open Questions

- Can we further reduce the approximation factor?
- For incremental: Lower bound of worst-case recourse k can be shown for < 2 -approximations.
- What about decremental setting?

Thank you!